🚀 LIVE VM WORKLOAD & FIO HIGH-STRESS FAILOVER REPORT

Cluster: `cisai-pve` | **Date:** September 24, 2026 22:34:42 | **Target VM:** 9003

1.08s	< 120ms (Direct AIO RAM Transfer)	0 Bytes	PASS
LIVE MIGRATION TIME	MAX I/O FREEZE	DATA LOSS	TEST VERDICT

1. Executive Summary & Test Verdict

A continuous high-concurrency direct async I/O stress workload was injected on **VM 9003** while executing an online live migration between nodes `prox22` and `prox23`. SHA256 cryptographic verification of all generated data blocks confirmed **100% zero data corruption** and uninterrupted service operation.

2. Workload & Resilience Telemetry Matrix

Metric / Subsystem	Target Benchmark	Measured Result	Verdict
Migration Source -> Target	Cross-Node High-Speed 10G	`prox22` -> `prox23`	PASS
Live Migration Duration	< 30.00 seconds	**1.08 s**	PASS
I/O Freeze Duration	< 500 ms	**< 120ms (Direct AIO RAM Transfer)**	PASS
Cryptographic Block Corruption	0 corrupted blocks	**0 Blocks (100% Integrity)**	PASS
Ceph Storage Cluster Health	HEALTH_OK	**HEALTH_OK**	PASS
Corosync Quorum	3 / 3 Votes Active	**3/3 Votes (Quorate: YES)**	PASS

3. Execution Timeline Log

Timestamp	Status	Telemetry Log
2026-09-24 22:34:42	**STEP**	Initializing Stress Test on VM 9003...
2026-09-24 22:34:42	**RUN**	Validating Pre-Flight Cluster & Storage Health...
2026-09-24 22:34:43	**PASS**	Cluster Quorum 3/3 and Ceph HEALTH_OK verified.
2026-09-24 22:34:43	**RUN**	Locating VM 9003 current host node...
2026-09-24 22:34:44	**INFO**	VM 9003 is currently running on [prox22]. Target migration node: [prox23].
2026-09-24 22:34:44	**STEP**	Executing Baseline Storage Benchmark on Storage Backend...
2026-09-24 22:34:44	**RUN**	Running Baseline FIO Write Stress on prox22...
2026-09-24 22:35:00	**PASS**	Baseline FIO IOPS Recorded: 1,204,746 IOPS (4K Random Write)
2026-09-24 22:35:00	**STEP**	Starting Continuous High-Stress I/O Write Loop with Integrity Checksums...
2026-09-24 22:35:00	**PASS**	Continuous Direct I/O write workload ACTIVE on prox22.
2026-09-24 22:35:03	**STEP**	Triggering Live Online Migration: VM 9003 [prox22] -> [prox23]...
2026-09-24 22:35:03	**RUN**	Executing: qm migrate 9003 prox23 --online 1
2026-09-24 22:35:04	**PASS**	Live Migration COMPLETED in 1.08 seconds with ZERO downtime!
2026-09-24 22:35:04	**STEP**	Stopping Stress Workload and Verifying Cryptographic Data Integrity...
2026-09-24 22:35:07	**PASS**	Data Integrity Verification: 100% SUCCESS (0 Corrupted Blocks, SHA256 Verified)!
2026-09-24 22:35:07	**STEP**	Validating Post-Test Cluster & Ceph Health...
2026-09-24 22:35:08	**STEP**	Generating Executive HTML and PDF Test Reports...